

JIYA ROSE JOHNSON

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Indian Institute of Technology Madras
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RESEARCH INTERESTS Potential Theory, Spectral Theory, Analytic Partial Differential Equations, Geometric Analysis

EMPLOYMENT Research Associate, Indian Institute of Technology Madras

EDUCATION **Indian Institute of Technology Madras**
PhD in Mathematics, Feb 2021-Jun 2026
Fields: Potential Theory, Analytic PDE
Thesis: Spectral properties of generalized logarithmic potential operators
Advisor: Dr. T. V. Anoop

Cochin University of Science and Technology
MSc in Mathematics, July 2018-June 2020
CGPA: 9.84 / 10

St. Joseph's College Irinjalakuda
Affiliated with the **University of Calicut**
BSc in Mathematics, Jul 2015-Apr 2018
CGPA: 5.74 / 6

PUBLICATIONS T. V. Anoop and Jiya Rose Johnson.
Reverse Faber-Krahn inequalities for the logarithmic potential operator.
Discrete and Continuous Dynamical Systems (2026): Vol.50, Pages 321-357.
[Doi: 10.3934/dcds.2026012](#).

PREPRINTS T. V. Anoop and Jiya Rose Johnson.
On the weighted logarithmic potential operator.
[arXiv:2602.18138](#).

Jiya Rose Johnson and Sheela Verma.
On eigenvalues of logarithmic potential operator in the hyperbolic space.
[arXiv:2601.20431](#).

AWARDS AND HONORS

Cochin University rank holder (Rank 2) in M.Sc. Mathematics

Calicut University rank holder (Rank 3) in B.Sc. Mathematics

Recipient of the title “Student of the Year,” awarded to the best outgoing degree student from St. Joseph’s College, Irinjalakuda (2018)

Recipient of the Prathibha Scholarship for pursuing a degree course in basic science, given by the Kerala State Council for Science, Technology and Environment to the top 100 students in Kerala (2015)

Summer Research Fellow, Indian Academy of Sciences (2019)

Advisor: Dr. Sheethal Dharmatti, IISER Thiruvananthapuram

CONFERENCE TALKS

On eigenvalues of logarithmic potential operator (Poster).

Geometry of Eigenvalues, Isaac Newton Institute for Mathematical Sciences, Cambridge, England. March 2026.

On eigenvalues of weighted logarithmic potential operator.

Seminars in PDE-III, IIT Madras. March 2026.

Faber–Krahn inequalities for the logarithmic potential operator.

UK Spectral Theory Workshop, University of Reading, England. August 2025.

Faber–Krahn type inequalities for the logarithmic potential operator.

Indian Women and Mathematics Annual Conference, IIT Patna. July 2025.

Reverse Faber–Krahn inequalities for the logarithmic potential operator.

Recent Advances in Nonlinear PDEs, IIT Madras. March 2025.

TEACHING EXPERIENCE

Teaching Assistantships, Indian Institute of Technology Madras

Series and Matrices (Jan–May 2022, 2023)

Multivariable Calculus (Jul–Nov 2022, 2023)

Complex Variables (Jan–May 2024)

Differential Equations (Jul–Nov 2024)

Teaching Assistantships, NPTEL

Measure and Integration (Jul–Oct 2023–2026)

Functional Analysis (Jan–Apr 2025–2026)

WORKSHOPS	<i>Symposium on Luis Caffarelli's Works</i>, TIFR-CAM Bengaluru, June 2023 <i>AIS on Elliptic and Parabolic PDEs</i>, IISc Bengaluru, December 2023
QUALIFIED EXAMS	CSIR NET in Mathematical Science December 2018 in the first attempt (All India Rank 51) CSIR JRF in Mathematical Science June 2019 (All India Rank 83) CSIR NET in Mathematical Science December 2019 (All India Rank 16) GATE 2020 (All India Rank 16)
ACADEMIC SERVICE	Delivered invited outreach talks at colleges across Kerala Conducted problem-solving sessions for competitive examinations Organized academic webinars during the COVID-19 pandemic Elected Vice Chairperson of the College Union, St. Joseph's College, Irinjalakuda
SKILLS	C, C++, PYTHON, SQL, Mathematica
REFERENCES	Dr. T. V. Anoop, Department of Mathematics, Indian Institute of Technology Madras, Email: anoop@iitm.ac.in Dr. Sheela Verma, Department of Mathematical Sciences Indian Institute of Technology (BHU), Varanasi, Email: sheela.mat@iitbhu.ac.in Dr. Anisa Chorwadwala, Department of Mathematics Indian Institute of Science Education and Research Pune, Email: anisa@iiserpune.ac.in